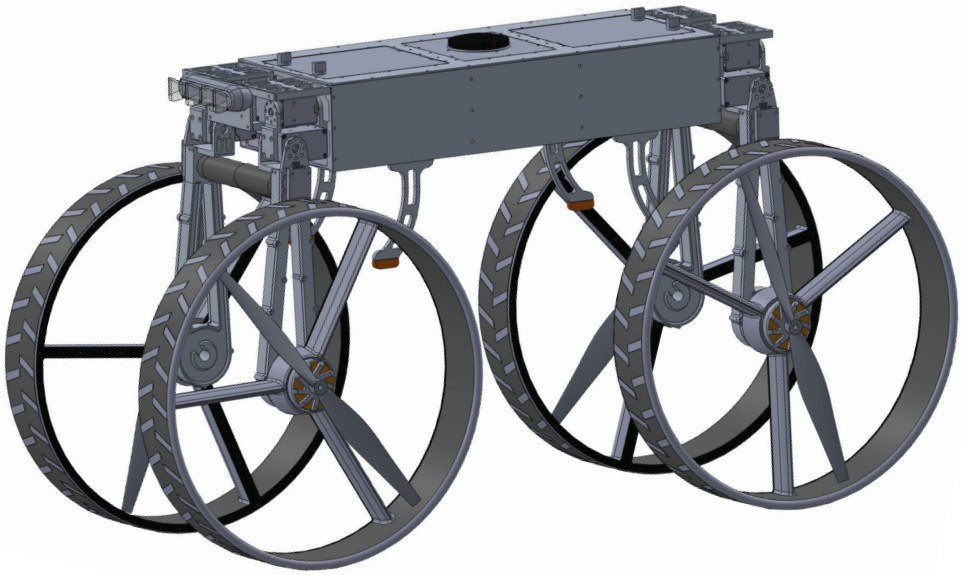
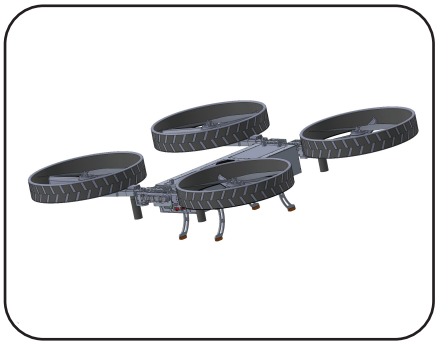
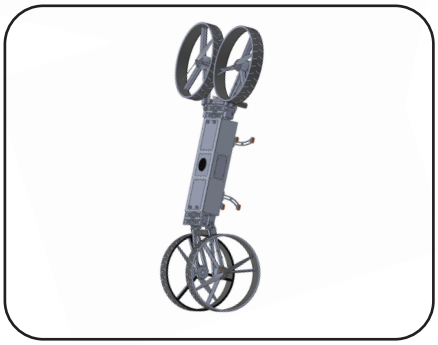
Caltech

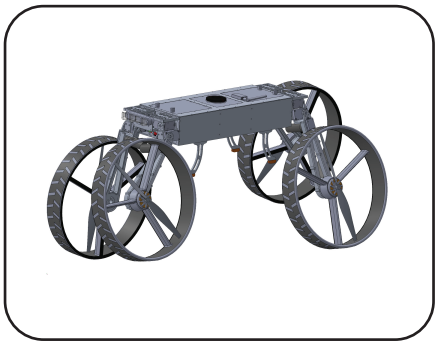
M4 v.3

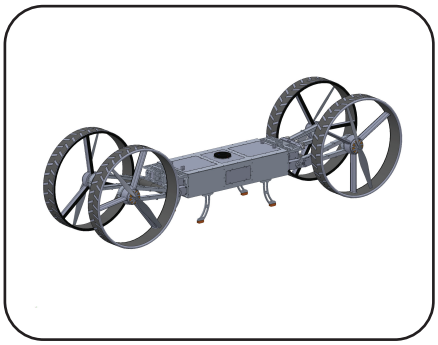
User Manual











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Using this manual

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* Safety and Legal Notice

* The M4 v.3 Robot must be operated in accordance with local laws and regulations.

* The M4 v.3 may not be operated or flown near or over people, roads, vehicles, buildings or anything that could result in damage or danger to people or property.

All missions must be validated and pass _____

Radio or video transmitters might require special licensing and/or permits.

Caltech and/ or its affiliates cannot be held responsible for vehicles that have not been operated or maintained in accordance with the guidelines set forth in this manual, or for behavior resulting from changes made to the vehicle.

* This operations manual is provided "As-Is". No rights can be obtained from the contents of this manual.

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1.0 Robot Details

Dimensions:

Wingspan: ___ cm
Length: ___ cm
Height: ___ cm
Payload bay __ x __ x __ cm

Weight and Payload:

Empty weight: ___ Kg
Empty weight including battery: __ Kg
Maximum takeoff weight: ___ Kg
Payload capacity: ___ Kg

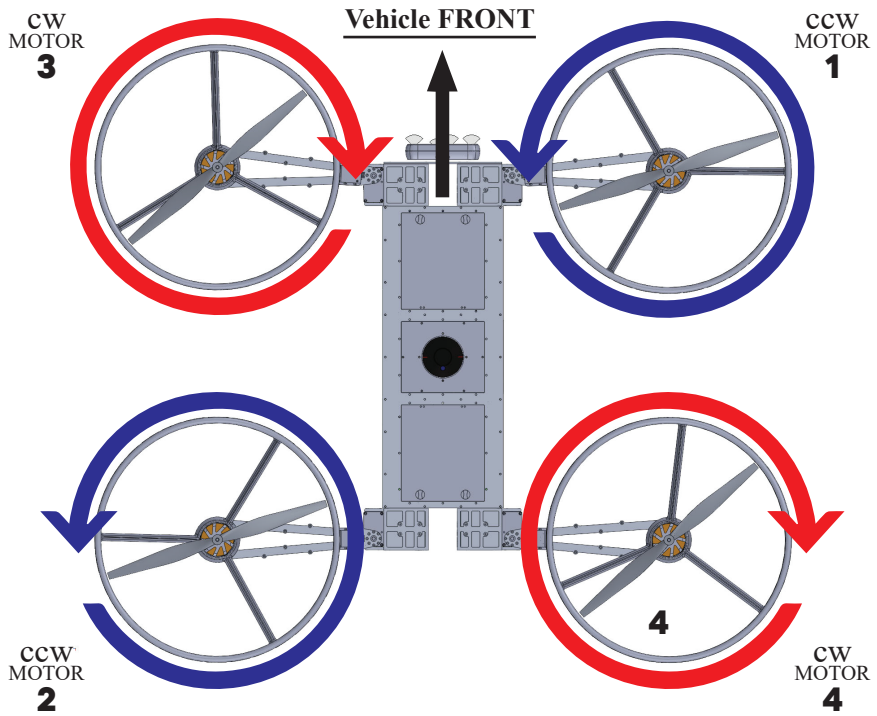
Flight Characteristics at no payload:

Cruise speed __ m/s (__ Km/h), (__ Mph)
Maximum speed __ m/s (__ Km/h), (__ Mph)
Stall speed __ m/s (__ Km/h), (__Mph)
Maximum flight time* __+ hours

Payload specifications:

Maximum payload weight (without Aux LiPo) 1.2Kg
Maximum payload weight (with Aux LiPo) 400g
Payload bay height: 80mm minimum (oval shaped)
Payload bay length: 120mm
Payload bay width: 200mm minimum (oval shaped)

Motor Configuration and Propellor Rotation



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1.1 Safety Features

Low Battery Failsafe Trigger

Default: Return at critical level, land at emergency level
Default Warn level: 10%
Default Failsafe level: 7%
Default Emergency level: 5%

These parameters define what the vehicle does when reaching low battery levels.

Note: The levels are those estimated to be reached when having flown to the landing site. This means that the further the vehicle is from its intended landing location, the sooner these actions will be taken. The vehicle will attempt to maintain these values when landed.

Warn level: The percentage where the vehicle will give a visible and audible warning to the Ground Control Station

Failsafe level: The level at which the vehicle initiates the critical battery action. (return)
Emergency level: The level at which the vehicle initiates the emergency battery action. (land)

RC Loss Failsafe trigger

Default: Disabled

Hold mode. If the Data Link can't be re-established the vehicle will remain in Hold Mode until the Low Battery Failsafe Trigger will be activated.
Default action: Warning
Default max radius: Disabled
Default max altitude: Disabled

The geofence failsafe trigger can be set to limit the vehicle's radius and/or altitude. When settings these parameters the vehicle will perform the defined action upon breaching any of these.

Return Home Settings

Default climb: 60m
Default home action: Land immediately

These settings define how the vehicle behaves when it engages the Return to Land function. The Climb altitude is the minimum altitude relative to home the vehicle will maintain for its return path. If the altitude of the M4 v.3 is lower than the Default climb value, in this case, 60 meters, the M4 v.3 will ascend to that Default climb altitude of 60 meters.
If the vehicle is higher at the point where the Return to Land is triggered it will maintain that altitude to return.
The M4 v.3 can perform an autonomous Return to Land when instructed from the _____ Station, when instructed from a mission, or when triggered by a failsafe event.

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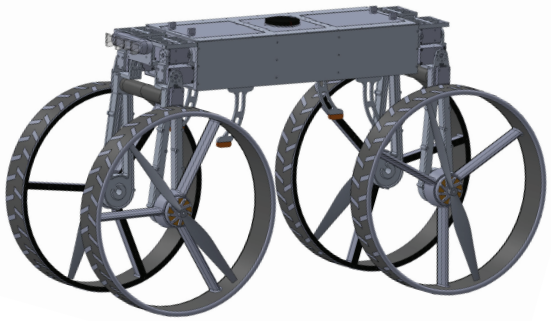
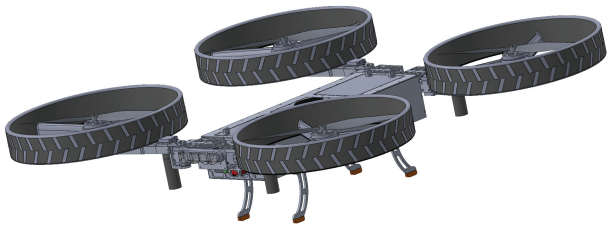
1.2 Modes of Oporation

MODE 1



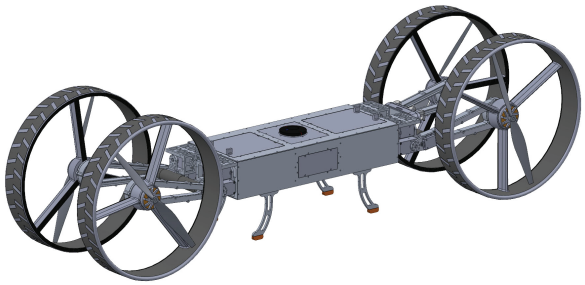
ROVER/ DRIVE MODE

MODE 2



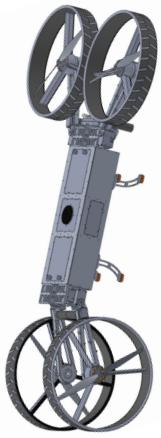
FLIGHT MODE

MODE 3



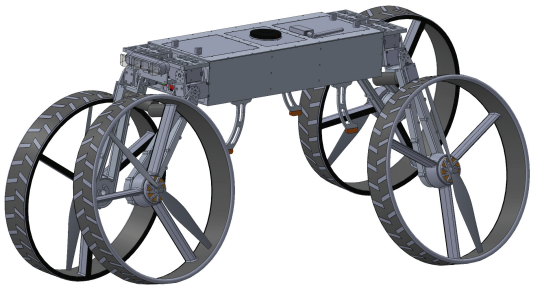
CROUCHING MODE

MODE 4



MEERKAT MODE

MODE 5



CREEP MODE

MODE 1
Rover/ Drive Mode

MODE 2
Flight Mode Empty weight: ___ Kg

MODE 3
Crouching Mode

MODE 4
Meercat Mode

MODE 5
Creep Mode

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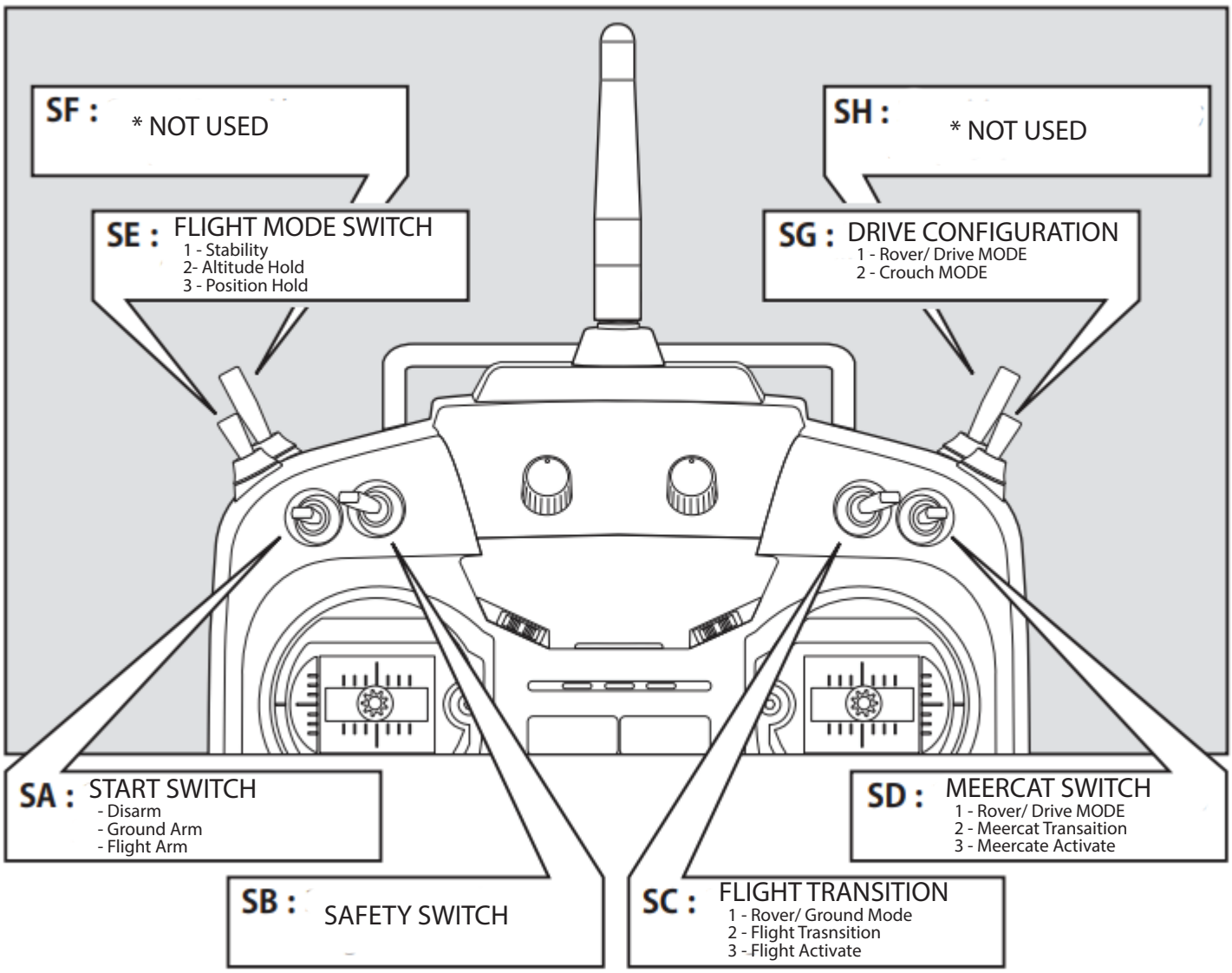
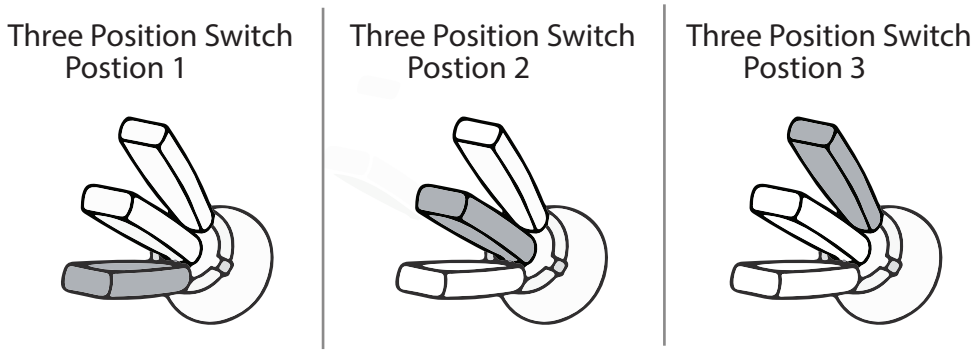
2.0 Transmitter Details

Model: FUTABA 18-SZ





** See Transmitter Manual for more information.

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3.0 Charger Specifications

Input Voltage:	90-264V AC
Dimensions:	15 x 12 x 7 inches
Weight (assembled):	approx. 15 pounds
Balance Accuracy:	10mV
Current Drain (Balancing):	1.2A per channel
Discharge Current Range	0.005-40A per channel (up to 70A in sync mode)
LiPo/LiIo/LiFe:	1-10S
Charge Power (Max):	1800W
Discharge (Max):	130W per channel (200W max)
External Discharge (Max):	1600W per channel
Cell Count (NiCd/NiMh):	1-25S
Pb Battery Cell Count:	1-18S
Charge Current Range:	0.005-40A per channel (up to 70A in sync mode)

** See Charger Manual for more information.

3.1 Battery Specifications

The M4 v.3 Uses two 10400mAh High Performance Lithium Batteries.

Nominal capacity: 10400mAh
Nominal voltage: 6S 22.2 volts
Maximum voltage: 25.2 volts
Minimum voltage: 18 volts
Recommended landing voltage: 21 volts
Chemistry: Lithium-Polymer
Maximum constant discharge: 455 amps
Maximum charge current: 52 amps
Watt hours: 230.9 Wh
Energy Density: 159 Wh/Kg
175C rating
138mm x 45mm x 111mm, 1456g

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3.2 Charging

Charging the LiPo:

The M4 v.3 includes a LiPo battery and charger.

This charger can charge an empty M4 v.3 LiPo battery in approximately ____ hours.

To charge the battery connect both the power and balance plugs in the charger, it will commence charging automatically.

When the LED light on the charger turns green the LiPo Batteries will have been fully charged.

PROCEEDURE:
1. Remove the battery from the vehicle. The battery should never be charged inside the vehicle.
2. Never charge batteries unattended unless you can charge them inside a LiPo safe.
3. Charge in an isolated area, away from other flammable materials.
4. Ensure the battery is at ambient temperature before charging.

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4.0 Preparation Before Flight

5.0 Operation Guide

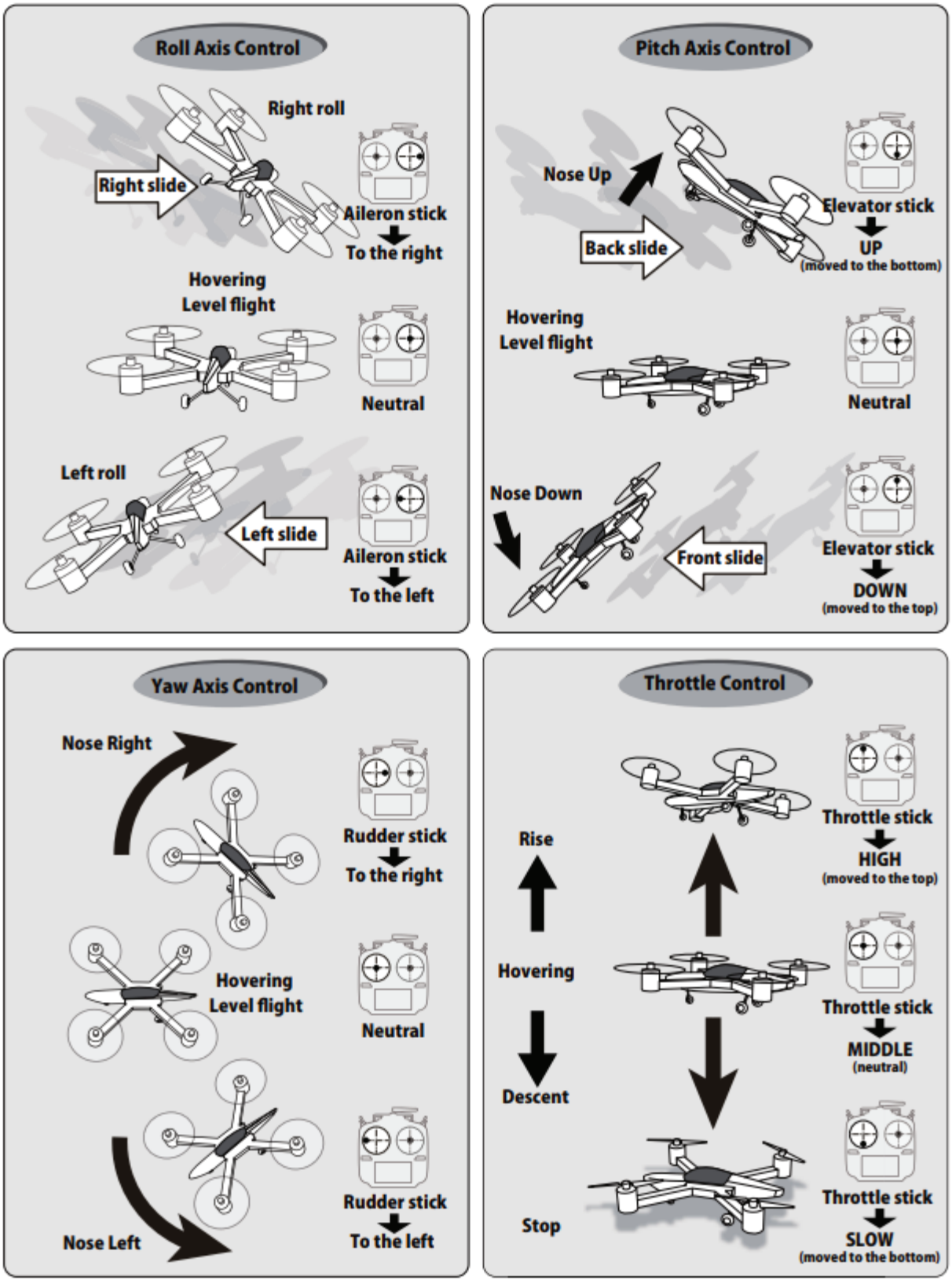
5.1 Paring

5.2 Initialization Detection

5.3 Compass Calibration

5.4 Ready to Fly

5.5 Flight Control



5.6 Return to Home (RTH)

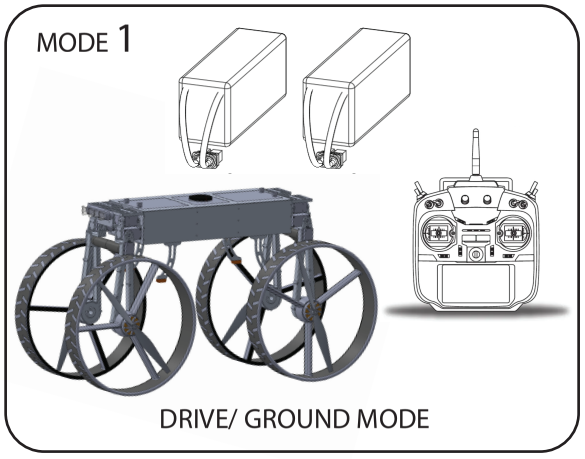
6.0 Maintenance

7.0 Trouble Shooting

Step 1

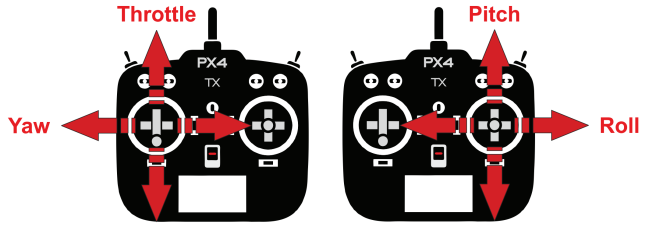
Start Up

- Perform Pre-Flight Checklist (see manual)
- Place M4 Robot on flat/ level surface
- Verify M4 Batteries are fully Charged
- Verfiy Controller Battery is fully Charged
- Turn on Controller/ Radio
- Verify MODEL select is set correctly in Controller/ Radio to M4 Robot
- Confirm all swtiches on Controller/ Radio are reset
- Check Propellers for tightness and condition
- Visually inspect Robot for flight worthiness
- Verify wheels are FREE from constraints
- Confirm Arms and Joints move freely
- Turn on Mission Planner
- Install Batteries into M4 and plug in
- Verify connection between Mission Planner and Controller
- Press button ontop of GPS dome to arm.



Step 2

Operation



Step 3

Post Operation

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M4 v.3

QUICK START GUIDE

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